

# Karsten Kirby Uy

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## Education

### University of British Columbia

Expected Graduation: Dec 2027

Bachelor of Applied Science in Computer Engineering (GPA: 88.8%)

Vancouver, BC

- **Relevant Coursework:** Digital Systems Design (SystemVerilog), Operating Systems (C), Signals and Systems, Machine Learning with Engineering Applications (Python), Software Construction I (Java), Signals and Systems (DSP)

## Experience

### Intel

May 2026 – Present

System Simulation Engineer Intern

Vancouver, BC

- Developed **SystemC** transaction-level models of hardware blocks, enabling **pre-silicon** firmware and software validation ahead of RTL and silicon availability.

### Synopsys Inc.

Jan 2025 – Aug 2025

Software Engineer Intern

Ottawa, ON

- Built a next-generation AI-driven EDA verification flow in the IP R&D team, designing Bash data pipelines (20K+ samples) to train ML models that increased verification coverage up to 8× and reduced runtime up to 4.5× across 7 VC Formal applications, integrating the flow into a C++ RTL feature extractor generating user-friendly TCL scripts.
- Engineered CLI and GUI workflows for an RTL analysis tool, streamlining 10+ verification and compilation workflows and integrating with hardware validation tools such as Timing Constraint Manager, VC Formal, and Fusion Compiler.
- Developed a comprehensive RTL validation framework consisting of 100+ test cases to quantify analyzer throughput and ensure functional parity during optimization.
- Streamlined multi-platform deployment by packaging ML inference engines into standalone binaries, enabling C++ integration and resolving dependency conflicts across legacy CentOS and modern AlmaLinux distributions.

### Insurance Corporation of British Columbia

May 2024 – Dec 2024

Software Developer Intern

North Vancouver, BC

- Developed and deployed a hybrid Python and Blue Prism RPA solution to automate multi-platform data transfers, reducing operational complexity and significantly lowering annual operational costs.
- Managed critical production issues for 30+ automation solutions by analyzing and resolving incidents to ensure high uptime, seamless integration, and productivity while conducting regression testing and verifying security patches.

## Projects

### FPGA Multi-Effects Audio Processor | GitHub | Demo | SystemVerilog, FPGA, Fixed-Point DSP, Python

- Engineered a synthesizable real-time audio DSP pipeline on the Intel Cyclone V (DE1-SoC): a 12-stage effect chain at 48 kHz/16-bit with sub-millisecond (<1ms) latency and an always-on YIN pitch tracker for chromatic tuning.
- Developed fixed-point DSP blocks including a 3rd-order polynomial  $\tanh$  distortion model, an 8-sample lookahead compressor, and a multiplier-free Feedback Delay Network reverb of 8 fractional-delay lines coupled by a lossless  $8 \times 8$  Hadamard matrix, holding 50 MHz timing closure through pipelining and multicycle-path constraints.
- Built a live JTAG-UART host link from an Avalon-MM adapter and a transport-agnostic command-parser FSM that lets a Python (Tkinter) GUI and CLI read, write, dump, and save a 512-byte parameter store on the running board.
- Architected power-persistent storage of 4 preset banks in EPCQ SPI flash via a custom erase/program FSM, with a gain-ramp fade FSM and GPIO/ADC pedal controls also mirrored to a class-compliant Arduino USB-MIDI controller.

### Parallel ARC4 Decryptor | SystemVerilog, FPGA, Cryptography

Nov 2025 - Dec 2025

- Engineered a synthesizable ARC4 engine on DE1-SoC, instantiating 150+ parallel cores to achieve a throughput of 3,500+ keys/sec for ciphertext cracking.
- Achieved timing closure at 120 MHz by pipelining RTL logic, optimizing resource utilization (92%), and mitigating routing congestion during the Quartus fitting stage.
- Validated hardware performance and functional correctness using ModelSim and SystemVerilog UVM-inspired testbenches to verify throughput and parallel execution.

## Technical Skills

**Design & RTL:** SystemVerilog, Verilog, C/C++, FSM Design, Fixed-Point DSP

**Verification & EDA:** SVA, Formal Verification (VC Formal), ModelSim, Functional Coverage

**Physical & Timing:** Static Timing Analysis (STA), Synthesis (Fusion Compiler), Quartus, FPGA Prototyping

**Scripting & Systems:** TCL, Python, Bash, Linux (Alma/CentOS), Git/Perforce, Bazel, SystemC